where a and b are arbitrary constants.

Differentiating equation (1) with respect to x, we get

$$\frac{2x}{a^2}+\frac{2y}{b^2}\frac{dy}{dx}=0 \Rightarrow \frac{x}{a^2}+\frac{y}{b^2}\frac{dy}{dx}=0 \qquad \ldots (2)$$

Differentiating equation (2) with respect to x, we get

$$\frac{1}{a^2}+\frac{1}{b^2}\left[y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right]=0 \Rightarrow \frac{1}{a^2}=-\frac{1}{b^2}\left[y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right]$$

Substituting the value of $\frac{1}{a^2}$ in equation (2) and simplifying, we get

$$-\frac{1}{b^2}\left[y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right]x+\frac{y}{b^2}\frac{dy}{dx}=0 \Rightarrow xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$$

which is the required differential equation.

Remark

The result of eliminating one arbitrary constant yields a first order differential equation and that of eliminating two arbitrary constants leads to a second order differential equation and so on.

EXERCISE 10.3

1. Find the differential equation of the family of (i) all non-vertical lines in a plane (ii) all non-horizontal lines in a plane.
2. Form the differential equation of all straight lines touching the circle $x^2+y^2=r^2$.
3. Find the differential equation of the family of circles passing through the origin and having their centres on the x-axis.
4. Find the differential equation of the family of all the parabolas with latus rectum $4a$ and whose axes are parallel to the x-axis.
5. Find the differential equation of the family of parabolas with vertex at $(0,-1)$ and having axis along the y-axis.
6. Find the differential equations of the family of all the ellipses having foci on the y-axis and centre at the origin.
7. Find the differential equation corresponding to the family of curves represented by the equation $y=\mathrm{A}e^{8x}+\mathrm{B}e^{-8x}$, where A and B are arbitrary constants.
8. Find the differential equation of the curve represented by $xy=ae^x+be^{-x}+x^2$.

Chapter 10 Differential Equation.indd 154 26-02-2020 17:24:01